1 RESEARCH/PROTOCOL TITLE

PREVALENCE AND INCIDENCE OF SEXUALLY TRANSMITTED INFECTIONS AND CONTRACEPTIVE USE AMONG FEMALE ADOLESCENTS IN TANZANIA - A PROSPECTIVE STUDY

Protocol Version and Date	Version 2.0, 13.11.2021
Funding Sponsor	University of Antwerp
Study Principle Investigator	
Study Site/area	Multi-centre: Dar es Salaam, Mwanza, Mbeya

RESEARCH SYNOPSIS

This is a prospective interventional study with quarterly visits for up to 12 months of follow up, 480 sexually active girls (by self-report), 16 – 20 years of age, willing to undergo study procedures and willing and able to provide independent informed consent/assent will be eligible for this study. Girls with known plans of relocation from study area for more than 3 consecutive months and with any medical condition that might make study participation unsafe or interfere with study objectives will be excluded.

Eligibility determination and enrollment into the study will be done on the same day (day 1), consenting/assenting (for <18 years) will be done, locator information collected and questionnaires will be administered to obtain demographic information. During subsequent follow up visits up to 12 months, questionnaires will be used to collect information on; sexual history and behavior, STI symptoms, use and attitudes of contraceptive. Urine will be collected to do a pregnancy test and Urine dipstick will be performed to detect urinary tract infections. Blood and genital swabs will also be collected for various laboratory tests such as HPV, HSV, Syphilis, BV, HBV serology etc.

Enrollment: The proposed duration for enrolling the 480 sexually active girls is approximately six months, which means that the last participant will exit the study six months after the first ever enrolled participant has exited the study.

RESEARCH AIMS/GOALS

1.1.1 To determine the prevalence of and factors associated with reproductive tract infections.

1.1.2 To determine the incidence of and factors associated with acquisition of reproductive tract infections.

1.1.3 To determine the prevalence of contraceptive use and factors associated with contraceptive use among adolescent girls living in informal settlements and other high population density low income settlements

1. List of Abbreviation

HPV	Human papilloma virus
STI	Sexually Transmitted Infection
SELIG	Screening and Enrollment Eligibility
SESX	Screening Enrollment Sexual history
ESLR	Enrollment Screening Lab Results
SEGE	Screening Enrollment Genital Exam
PENR	Participant Enrollment
FSX	Follow up sexual history
FGE	Follow up Genital Exam
HBV	Hepatitis B virus
PTID	Participant ID
MV	Missed Visit

Screening and Enrollment Eligibility Form (SELIG)

Screening ID:	**Participant ID:**	**Visit Date:**
☐☐-☐☐☐☐☐	☐☐-☐☐☐-☐	☐☐ ☐☐☐ ☐☐☐☐

1.	1a. Did the participant provide informed consent? Yes ☐ No ☐ → ***Go to item 2*** 1b. Date when informed consent for screening was marked or signed? ☐☐ ☐☐☐ ☐☐☐☐ ***(Then go to item 4)*** dd mmm yyyy
2.	2a. Was the participant's guardian/parent willing and able to provide independent, written, informed consent for screening? Yes ☐ No ☐ → ***Ineligible(End of Form)*** 2b. Date when informed consent for screening was marked or signed? ☐☐ ☐☐☐ ☐☐☐☐
3.	3a. Was the participant willing and able to provide independent, written, informed assent for screening? Yes ☐ No ☐ → ***Ineligible(End of Form)*** 3b. Date when informed assent for screening was marked or signed? ☐☐ ☐☐☐ ☐☐☐☐ dd mmm yyyy

Question 4-7 should be answered 'yes' for the participant to be eligible

4.	Is the participant 16 – 20 years?	Yes ☐ No ☐
5.	Is the participant/ participant's guardian/parent willi ng and able to provide adequate locator information?	Yes ☐ No ☐
6.	Is the participant willing and able to undergo genital examinations?	Yes ☐ No ☐
7.	Is the participant willing to remain in follow up for 3 years?	Yes ☐ No ☐

Exclusion Criteria *Questions 8-11 should be answered 'no' for the participant to be eligible*

8.	Has the participant ever had sex?	Yes ☐ No ☐ → Go to item 9
	8a. if yes in item 8 specify number of life time partne rs ☐☐	(if >1 participant ineligible)
9.	Does the participant have any condition that, in the opinion of the Site Investigator or designee, makes participation in the study unsafe, complicates interpretation of the study outcome data, or otherwise interferes with achieving the study objectives?	Yes ☐ No ☐

If participant is still eligible continue with blood draws. If ineligible at this point, end form.

10.	Is the participant HIV positive?	Yes ☐ No ☐
11.	Is the participant HSV 2 positive?	Yes ☐ No ☐
12.	Based on the information from item 1-11, is the participant eligible for enrollment?	☐ Eligible ☐ Not eligible

If participant is eligible but does not enroll, continue, otherwise end of form.

13.	Reason eligible participant did not enroll: ☐ Declined Enrolment ☐ Investigator decision ☐ Did not return for enrollment ☐ Other Specify_____

Completed by (Initials/date) _____

Screening and Enrollment Genital Exam page 1 (SEGE)

Participant ID: ☐☐ ☐☐☐ ☐	**Visit Date:** ☐☐ ☐☐☐ ☐☐☐☐ *dd mmm yyyy*
1.	Last Menstrual Period ☐☐ ☐☐☐ ☐☐☐☐ *dd mmm yyyy*
2.	a. Was a genital exam done? Yes No b. If genital exam was not done, please provide a reason _____
3.	Is the participant circumcised? Yes No N/A
4.	Were any of the following signs noted? a. Redness Yes No N/A b. Abnormal vaginal discharge Yes No N/A c. Swelling Yes No N/A d. Ulcers Yes No N/A Location of ulcers: Mons, Perianal, Labia, Buttock, Perineal, Leg/hip/back e. Other Yes No N/A Specify: _____

Completed by (initial/date) _____

Participant ID: ☐☐ ☐☐☐ ☐		**Visit Date:** ☐☐ ☐☐☐ ☐☐☐☐ *dd mmm yyyy*		
5.	Was a HSV-PCR swab collected?	☐ Yes	☐ No	☐ N/A
6.	Was a vaginal swab for BV gram stain collected?	☐ Yes	☐ No	☐ N/A
7.	Was a vaginal swab for BV storage collected?	☐ Yes	☐ No	☐ N/A
8.	Was a vaginal swab for GC, CT, TV collected?	☐ Yes	☐ No	☐ N/A
9.	Was cervicovaginal lavage done?	☐ Yes	☐ No	☐ N/A
10.	Was an archive sample for HSV testing taken?	☐ Yes	☐ No	
11.	Was a host DNA sample taken?	☐ Yes	☐ No	
	Etiologic Diagnosis			
12.	Laboratory tests done for etiological diagnosis of STD s? ☐ Not done → (***end of form***)			
	a. Gonorrhea: ☐ not done ☐ done → Result: ☐ negative ☐ positive			
	b. Chlamydia: ☐ not done ☐ done → Result: ☐ negative ☐ positive			
	c. Trichomoniasis: ☐ not done ☐ done → Result: ☐ negative ☐ positive			
	d. Bacterial Vaginosis: ☐ not done ☐ done → Result: ☐ negative ☐ positive			

Completed by (initials/date) _____

PREVALENCE AND INCIDENCE OF SEXUALLY TRANSMITTED INFECTIONS AND CONTRACEPTIVE USE AMONG FEMALE ADOLESCENTS IN TANZANIA - A PROSPECTIVE STUDY

Participant ID: ☐☐ - ☐☐☐ - ☐ **Date:** ☐☐ ☐☐☐ ☐☐☐☐ *dd mmm yyyy* **Visit code:** ☐☐ ☐

Participant Follow-up Sexual History and Behavior (FSX)

CRF not administered ☐

Participant Follow-up Sexual History and Behavior *This is an interviewer administered form.* **Page 1**

Now I'm going to ask you about your sexual practices. While some of the information may be embarrassing or difficult to remember, please try to give your best answers and be as honest as you can. Whenever sex or sexual intercourse is stated, it includes vaginal and anal sex, but not oral sex. The term 'condom' refers to either a male or a female condom.

1.

a. Have you ever had sex? ☐ Yes No → ***End of form***

b. When did you first have sex? ☐☐ ☐☐☐ ☐☐☐☐ *dd mmm yyyy* **OR** Already reported at visit ☐☐.☐ → Go to item 2

c. Was a condom used the first time you had sex? ☐ Yes ☐ No

d. Was the sexual act your choice and with your agreement? ☐ Yes ☐ No

e. Were you offered a reward/ money for sex? ☐ Yes ☐ No

f. How old was the person you first had sex with? ☐☐ *Years*

g. How long had you known the person before having sex? ☐☐ ☐ Days ☐ Months ☐ Years

2.

a. In the past 3 months, how many times have you had sex? times → **If 00, go to item 3**

b. How many times did you use a condom? ☐☐ times

c. How many sexual partners have you had in the past 3 months? ☐☐

d. How many new sexual partners have you had in the past 3 months? ☐☐

Completed by: (initials/date) _____

Participant ID: ☐☐ - ☐☐☐ - ☐ **Date:** ☐☐ ☐☐☐ ☐☐☐☐ **Visit code:** ☐☐ ☐
dd mmm yyyy

Participant Follow-up Sexual History and Behavior (FSX)

CRF not administered ☐

Participant Follow-up Sexual History and Behavior **Page 2**

3.

a. In the past 3 months have you had sex? ☐ Yes No → ***Go to item 4***

b. How many times did you have sex? ☐☐ times

c. How many times did you use a condom? times

4.

a. In the past three months, have you used any method of contraception? ☐ Yes ☐ No → ***End of form***

b. Which method do/did you use? *Mark all that apply*

☐ Oral ☐ Implant ☐ Safe days
☐ IUD ☐ Emergency pills ☐ Other,(*specify*) _____
☐ Injectable ☐ Condoms

Completed by: (*initials/date*)_____

Participant ID: ☐☐ - ☐☐☐ - ☐ **Date:** ☐☐ ☐☐☐ ☐☐☐☐ *dd mmm yyyy* **Visit code:** ☐☐ ☐

Participant Follow-up Genital Exam (FGE)

CRF not administered ☐

Participant Follow-up Genital Exam	**Page 1**
1.	a. Date of Last menstrual Period ☐☐ ☐☐☐ ☐☐☐☐ *dd mmm yyyy* b. Was a genital exam done? ☐ Yes ☐ No (if no complete item 1 c then skip to 4g) *c.* If a genital exam was not done, please provide a reason:_____
2.	i. Were any of the following abnormalities noted on external genital examination? a. Abnormal vaginal discharge ☐ Yes ☐ No ☐ Not examined b. Genital warts ☐ Yes ☐ No ☐ Not examined c. Other: ☐ Yes ☐ No ☐ Not examined Specify: _____ ii. Was a speculum exam done? ☐ Yes ☐ No ☐ N/A → **Go to item 3** On speculum exam were any of the following abnormalities noted? a. Cervical mucopus ☐ Yes ☐ No ☐ Not examined b. Other: ☐ Yes ☐ No ☐ Not examined Specify: _____
3.	Does the participant have genital lesions consistent with herpes? ☐ Yes → ***Complete SGHE Form and collect HSV culture swab*** ☐ No ☐ Not examined
4.	a. Was a HSV-PCR swab collected? ☐ Yes ☐ No ☐ N/A b. Was a vaginal swab for BV gram staining collected? ☐ Yes ☐ No ☐ N/A c. Was a vaginal swab for BV storage collected? ☐ Yes ☐ No ☐ N/A d. Was a vaginal swab for GC, TV and Chlamydia collected? ☐ Yes ☐ No ☐ N/A e. Was a cervicovaginal lavage done? ☐ Yes ☐ No ☐ N/A f. Was a HSV culture swab collected? ☐ Yes ☐ No ☐ N/A g. Was a HIV/HSV ELISA sample collected? ☐ Yes ☐ No

Completed by (Initials/date**)**_____

Participant ID: ☐☐ - ☐☐☐ -☐ **Date:** ☐☐ ☐☐☐ ☐☐☐☐ **Visit code:** ☐☐ . ☐
dd mmm yyyy

Participant Follow-up Genital Exam (FGE)

Participant Follow-up Genital Exam	**Page 2**
Etiologic Diagnosis	
5.	Laboratory tests done for etiological diagnosis of STDs? Not done (skip to 6) a. Gonorrhea: ☐ not done ☐ done → Result: ☐ negative ☐ positive b. Chlamydia: ☐ not done ☐ done → Result: ☐ negative ☐ positive c. Trichomonas: ☐ not done ☐ done → Result: ☐ negative ☐ positive d. Bacterial Vaginosis: ☐ not done ☐ done → Result: ☐ negative ☐ positive
6.	Since the last visit, has the participant sero-converted to HSV? ☐ Yes No → ***Go to item 8*** **6a.** Date sero-conversion confirmed: ☐☐ (dd) ☐☐☐ (mmm) ☐☐☐☐ (yyyy) **OR** already reported at visit: ☐☐ . ☐
7.	Is participant eligible for HSV shedding sample collection at home? ☐ Yes ☐ No N/A → ***Go to item 8*** **7a**. Was participant trained on self-collection of genital swabs? ☐ Yes ☐ No If "No" specify the reason **then go to item 8** _____ **7b**. Was participant provided with a 2 week self-collection home kit and given a return date for returning the swabs? ☐ Yes ☐ No → **Complete PSIS Form** If "No" specify the reason _

Completed by (Initials/date**):** _____

PREVALENCE AND INCIDENCE OF SEXUALLY TRANSMITTED INFECTIONS AND CONTRACEPTIVE USE AMONG FEMALE ADOLESCENTS IN TANZANIA - A PROSPECTIVE STUDY

Missed Visit (MV)

PARTICIPANT ID ☐☐ ☐☐☐ ☐

Visit Month ☐☐ ☐

Form Completion ☐☐ (dd) ☐☐☐ (mmm) ☐☐ (yy)

Missed Visit

Instructions: *Complete this form when the window has closed for the visit.*
For Visit Month, enter the visit code of the scheduled visit that was missed.

1

Reason visit missed:

- ☐ School commitments
- ☐ Work issues
- ☐ Parental refusal
- ☐ Travelling or out of area
- ☐ Illness or hospitalized
- ☐ Transport/fare issues
- ☐ Refused visit
- ☐ Relocated or moved
- ☐ Unknown/unable to contact
- ☐ Incarcerated
- ☐ Other, specify _____

2 **Comments:**

Version 2.0, 13-Nov-2021	**English**	Completed by:_____(initials/date)